

Protocols for Creating Anyons and Defects via Gauging

Anasuya Lyons^{1,*}, Chiu Fan Bowen Lo,¹ Nathanan Tantivasadakarn², Ashvin Vishwanath^{1b}, and Ruben Verresen^{3,1}

¹*Department of Physics, Harvard University, Cambridge, Massachusetts 02138, USA*

²*Walter Burke Institute for Theoretical Physics and Department of Physics, California Institute of Technology, Pasadena, California 91125, USA*

³*Pritzker School of Molecular Engineering, University of Chicago, Chicago, Illinois 60637, USA*

(Received 3 January 2025; revised 22 August 2025; accepted 17 October 2025; published 13 November 2025)

Creating and manipulating anyons and symmetry defects in topological phases, especially those with a non-Abelian character, constitutes a primitive for topological quantum computation. We provide a physical protocol for implementing the ribbon operators of non-Abelian anyons and symmetry defects. We utilize dualities, in particular, the Kramers-Wannier or gauging map, which have previously been used to construct topologically ordered ground states by relating them to simpler states. In this Letter, ribbon operators are implemented by applying a gauging procedure to a lower-dimensional region of such states. This protocol uses sequential unitary circuits or, in certain cases, constant-depth adaptive circuits. We showcase this for anyons and defects in the $\mathbb{Z}_3$ toric code and S_3 quantum double. The general applicability of our method is demonstrated by deriving unitary expressions for ribbon operators of various (twisted) quantum doubles.

DOI: 10.1103/q2c6-nd6r

Introduction—Duality transformations are able to map complicated theories to simpler ones. Case in point is the Kramers-Wannier transformation [1–20], which maps the topologically ordered toric code [21] to a trivial paramagnet. Its group-based generalizations similarly relate product states to any quantum double $D(G)$ [7,22]. These represent deconfined phases of G gauge theories [21] whose anyonic quasiparticles exhibit generalized exchange statistics [23–27], implying topological degeneracy. Recently, the duality maps for solvable groups G have been rewritten as finite-depth measurement-based circuits, allowing for efficient preparation of a variety of interesting topological phases [28–35]. For instance, such a measurement-based scheme has been used to create D_4 topological order in the laboratory [36].

Here, we use dualities to construct ribbon operators [21,37,38] for topological defects [39–46] and anyons. The main conceptual result is illustrated in Fig. 1: we first map a one-dimensional region of a topological order to a simpler theory, then create a precursor to a given topological excitation. By reapplying the duality map, we return to the original topological order, which now contains the desired excitation. We note that Ref. [33] also used the interplay between gauging and anyon operators to diagnose certain symmetry-enriched topological orders.

We first exemplify our method for the $\mathbb{Z}_3$ toric code, then generalize to any quantum double $D(G)$. Finally, we discuss the applicability of our protocols for understanding “twisted” quantum double models [47–50]. In contrast

to the existing literature on quantum double ribbons [21,37,38], our approach is both conceptually and algebraically simple, relying solely on the properties of the G gauging map [22,28]. This simplicity underscores its practicality, as we can easily obtain efficient circuits for non-Abelian excitations ideal for near-term quantum devices. Indeed, in a companion paper using a trapped-ion quantum processor [51], we implement the $\mathbb{Z}_3$ toric code symmetry defect circuits introduced in the present Letter. Adaptive, efficient circuits for the creation of anyons in quantum doubles of solvable groups have been proposed before [31]; our approach pinpoints the gauging and ungauging layers inherent in the ribbon structure as the steps able to be made finite depth. This conceptual insight (i) yields intuitive

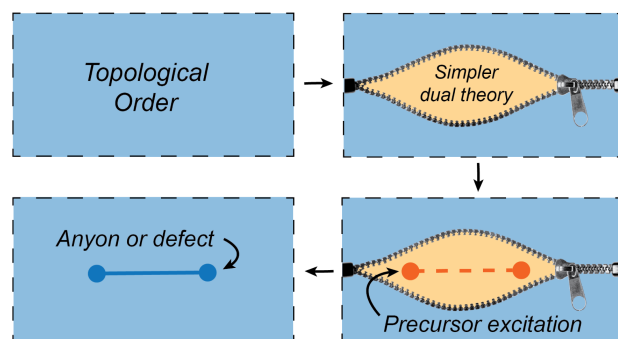


FIG. 1. Taking advantage of dualities: To create a given anyon or defect in a topological order, we apply a duality mapping in a finite region, “unzipping” to the simpler dual theory. Then, we create a “precursor” excitation, typically with a simple finite-depth circuit. Finally, we “zip” back up to the original theory, leaving behind the desired excitation.

*Contact author: anasuya_lyons@g.harvard.edu

unitary ribbon operators for *any* group-based non-Abelian anyon, (ii) gives a systematic construction for a wide range of “symmetry defect” ribbons, and (iii) opens up the possibility of constructing efficient ribbon operators for topological orders outside of the Kitaev quantum double.

$\mathbb{Z}_3$ toric code symmetry defects—We first discuss how to create symmetry defects in the $\mathbb{Z}_3$ toric code [21]. The computational qutrit basis is labeled $|n\rangle$, with $n = 0, 1, 2$. Define the qutrit clock and shift operators,

$$\begin{aligned} \mathcal{Z}|n\rangle &= \omega^n|n\rangle, & \mathcal{X}|n\rangle &= |n+1\rangle, \\ \mathcal{Z}^2 &= \mathcal{Z}^\dagger, & \mathcal{X}^2 &= \mathcal{X}^\dagger, & \mathcal{Z}\mathcal{X} &= \omega\mathcal{X}\mathcal{Z}, \end{aligned} \quad (1)$$

with $\omega = e^{i(2\pi/3)}$. The Hamiltonian is given by $H_{\mathbb{Z}_3} = -\sum_v A^{(v)} - \sum_p B^{(p)} + \text{H.c.}$, where [52]

$$A^{(v)} = \mathcal{X}^\dagger \mathcal{X} \quad B_p = \mathcal{Z}^\dagger \mathcal{Z} \quad (2)$$

A note on notation: throughout, any sub- or superscripts in parentheses denote a physical index. Violations of the vertex stabilizers are known as e (ω eigenvalue) and e^2 (ω^* eigenvalue) anyons, while plaquette violations are m and m^2 , respectively.

The $\mathbb{Z}_3$ toric code has a charge-conjugation symmetry permuting $e \leftrightarrow e^2$ and $m \leftrightarrow m^2$, not present in its $\mathbb{Z}_2$ counterpart [43,44,46]. A $\mathbb{Z}_3$ anyon encircling a charge-conjugation defect is transformed to its conjugate particle. Thus, these defects are non-Abelian, as they can absorb and emit Abelian particles [40,41]. The charge-conjugation symmetry is realized on site by the charge-conjugation operator $\mathcal{C}$,

$$\mathcal{C}|n\rangle = |-n \bmod 3\rangle \Rightarrow \mathcal{C}\mathcal{X}\mathcal{C} = \mathcal{X}^\dagger, \quad \mathcal{C}\mathcal{Z}\mathcal{C} = \mathcal{Z}^\dagger. \quad (3)$$

A closed charge-conjugation boundary is created by acting $\prod \mathcal{C}$ in a finite region of the lattice. Since inside this region, $\mathcal{C}A^{(v)}\mathcal{C} = A^{(v)\dagger}$ and $\mathcal{C}B^{(p)}\mathcal{C} = B^{(p)\dagger}$, the Hamiltonian only changes near the boundary of this region,

$$\begin{aligned} \tilde{A}^{(v)} &= \mathcal{X}^\dagger \mathcal{X} & \tilde{B}^{(p)} &= \mathcal{Z}^\dagger \mathcal{Z} \end{aligned} \quad (4)$$

Any anyon string that crosses the defect line must transform to its conjugate or violate these modified stabilizers. To create point charge-conjugation defects, we must terminate the defect line. This is not possible using a bulk symmetry membrane—we need a circuit localized to the defect line itself. We will show that such a circuit exists, whose form we will derive and unpack,

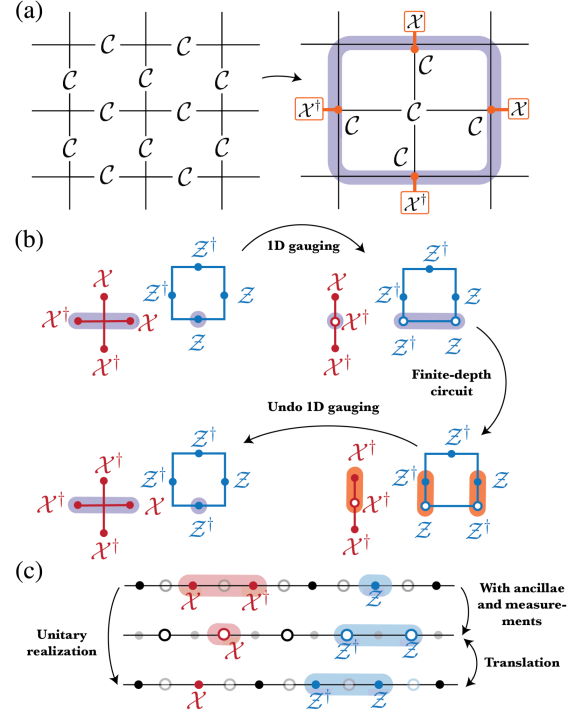


FIG. 2. Creating a charge-conjugation defect in $\mathbb{Z}_3$ toric code: (a) A charge-conjugation symmetry membrane reduces to a boundary circuit of controlled- $\mathcal{X}$ gates by using variables of the pregauged theory [see Eq. (9)]. (b) To create charge-conjugation defects: Map back to the necessary vertices using the 1D gauging map (affected degrees of freedom are highlighted in purple), then apply the derived boundary action (support highlighted in orange). Finally, map back to edges. (c) The action of the measurement-based and unitary realizations of $\text{KW}_{\mathbb{Z}_3}^{\text{1D}}$ differ by half a lattice translation. Here, solid circles mark edges, open circles mark vertices.

$$U_{\text{c.c.}}^\gamma = \text{KW}_{\mathbb{Z}_3}^{\text{1D};\gamma} \left(\prod_{v \in \gamma} \mathcal{C} \mathcal{X}^{\eta(e_v)} \right) \text{KW}_{\mathbb{Z}_3}^{\text{1D};\gamma^\dagger}; \quad \begin{matrix} e_v \\ -v \end{matrix} \quad (5)$$

$$\mathcal{X} \mathcal{X}^\dagger \xrightarrow{\text{KW}_{\mathbb{Z}_3}^{\text{1D}}} \mathcal{X} \quad \mathcal{Z} \xrightarrow{\text{KW}_{\mathbb{Z}_3}^{\text{1D}}} \mathcal{Z}^\dagger \mathcal{Z}$$

The action of $U_{\text{c.c.}}^\gamma$ is represented in Figs. 2(a) and 2(b). $\text{KW}_{\mathbb{Z}_3}^{\text{1D};\gamma^\dagger}$ is the one-dimensional (1D) (un)gauging map along an open path γ . This map may be more familiar as the Kramers-Wannier (KW) duality of the 1D Ising model: it maps the ferromagnetic phase (which we can think of as living on the edges along γ) to the paramagnetic phase (which lives on the vertices along γ) and vice versa. This duality is a result of gauging the $\prod X$ symmetry of the Ising paramagnet. The definition of the mapping given in the second line of Eq. (5) is the $\mathbb{Z}_3$ analog. We can roughly think of our topological state, restricted to a line, as being the ferromagnet—we need this 1D mapping to access its dual paramagnet, where the action of the symmetry is simpler. Note that we write $\text{KW}_{\mathbb{Z}_3}^{\text{1D}}$ rather than an explicit

circuit, as both measurement-based [28] and unitary [19,53] realizations of the same mapping exist and simply differ by half a lattice translation—see Fig. 2(c) for the action of the 1D gauging map in both cases [54]. The unitary implementation of $\text{KW}_{\mathbb{Z}_3}^{\text{1D}}$ is a sequential circuit of qutrit controlled-NOT (CNOT) gates: $\prod_{i \in \gamma} C\mathcal{X}_{i \rightarrow i+1}$.

The middle portion of $U_{\text{c.c.}}^{\gamma}$ acts between the uncovered vertex degrees of freedom and their neighboring edges perpendicular to γ , with $C\mathcal{X}|n, m\rangle = |n, m + n(\bmod 3)\rangle$. Here, $\eta(e_v) = \pm 1$ is some sign structure depending on the lattice, to be specified later. This part of the circuit implements the charge-conjugation symmetry on the hybrid state created by the first application of $\text{KW}_{\mathbb{Z}_3}^{\text{1D}}$.

To derive and explain Eq. (5), note that the $\mathbb{Z}_3$ paramagnet $\prod_v (|0\rangle + |1\rangle + |2\rangle)$ has the global $\mathbb{Z}_2$ symmetry $\prod \mathcal{C}$, along with its $\mathbb{Z}_3$ symmetry $\prod \mathcal{X}$. Upon gauging the latter using the 2D gauging map [defined in Eq. (6) below] the $\mathbb{Z}_3$ paramagnet is mapped to the $\mathbb{Z}_3$ toric code. Charge conjugation acts trivially in a finite region of the $\mathbb{Z}_3$ paramagnet—therefore, all the key properties of the charge-conjugation symmetry membrane on the $\mathbb{Z}_3$ toric code are captured by the 2D gauging map itself. By “pushing” the membrane from the $\mathbb{Z}_3$ toric code through the $\mathbb{Z}_3$ gauging map back to the paramagnet, we can simplify the bulk operator to some action purely at the membrane boundary.

More precisely, we define $\text{KW}_{\mathbb{Z}_3}^{2\text{D}}$ as the map that takes vertices to edges in the following way:

$$\begin{array}{ccc} \text{---}\mathcal{X}\text{---} & \xrightarrow{\text{KW}_{\mathbb{Z}_3}^{2\text{D}}} & \begin{array}{c} \mathcal{X}^\dagger \\ | \\ \mathcal{X}^\dagger \end{array} \\ & & \begin{array}{c} \mathcal{Z}^\dagger \\ | \\ \mathcal{Z} \end{array} \end{array} \quad \begin{array}{ccc} \begin{array}{c} \mathcal{Z}^\dagger \\ | \\ \mathcal{Z} \end{array} & \xrightarrow{\text{KW}_{\mathbb{Z}_3}^{2\text{D}}} & \text{---}\mathcal{Z}\text{---} \end{array} \quad (6)$$

The above fully specifies how $\text{KW}_{\mathbb{Z}_3}^{2\text{D}}$ transforms $\mathcal{C}$, as

$$\mathcal{C} = |0\rangle\langle 0| + \mathcal{X}|1\rangle\langle 1| + \mathcal{X}^\dagger|2\rangle\langle 2| = \sum_{n=0}^2 \mathcal{X}^n T_n, \quad (7)$$

where $T_n = |n\rangle\langle n| = \frac{1}{3}(1 + \omega^{-n}\mathcal{Z} + \omega^n\mathcal{Z}^\dagger)$. As a single vertex $\mathcal{Z}$ is not symmetric, we leave it behind when pushing $\mathcal{C}^{(\text{vertex})}$ through the gauging map. Similarly, we leave behind the $\mathcal{X}$ operators involved in $\mathcal{C}^{(\text{edge})}$,

$$\begin{aligned} \text{KW}_{\mathbb{Z}_3}^{2\text{D}} \cdot \text{---}\mathcal{C}\text{---} &= \sum_{n=0}^2 (A^{(v)})^n \cdot \text{KW}_{\mathbb{Z}_3}^{2\text{D}} \cdot \text{---}T_n\text{---} \\ \text{---}\mathcal{C}\text{---} \cdot \text{KW}_{\mathbb{Z}_3}^{2\text{D}} &= \sum_{n,n'} \mathcal{X}^{n-n'} \cdot \text{KW}_{\mathbb{Z}_3}^{2\text{D}} \cdot \begin{array}{c} T_{n'} \\ | \\ T_n \end{array} \end{aligned} \quad (8)$$

Thus, applying $\mathcal{C}$ to a pregauged vertex v is equivalent to applying $A^{(v)}$ after the gauging map, controlled on v . Acting $\mathcal{C}$ on an edge is equivalent to a product of $C\mathcal{X}$, controlled on the neighboring vertices.

Our first key observation is that Eq. (8) allows us to write down a purely one-dimensional operator that creates a charge-conjugation boundary [see Fig. 2(a)],

$$\prod_{v \in \gamma} C\mathcal{X}_{(v \rightarrow e_v)}^{\eta(e_v)}, \quad (9)$$

where γ is a closed path, e_v is defined as in Eq. (5), and $\eta(e) = +1(-1)$ if e_v points south, west (north, east).

We can create point defects by acting with a finite segment of this string operator. Starting from the toric code, however, we naively only have access to the edges, while Eq. (9) requires we use vertices as control qutrits. Our second key observation is that we can access the necessary vertices by simply ungauging the toric code along a line. We can interpret this as gauging a 1-form symmetry [55,56]: along the line, the 1-form symmetry looks like a 0-form symmetry of a 1D system. This means we can use $\text{KW}_{\mathbb{Z}_3}^{\text{1D}}$ to gauge it, allowing simultaneous access to vertex and edge degrees of freedom.

We have arrived at a 1D protocol to create charge-conjugation defects: (1) Access the vertices needed to act as controls by applying the 1D gauging map along the desired path. (2) Apply $C\mathcal{X}$ between the (ungauged) vertices along the line and the (gauged) edges just outside. (3) Undo the 1D gauging, mapping back to edge degrees of freedom everywhere. Combining all steps into a single unitary yields Eq. (5). Figure 2(b) shows that this circuit gives exactly the same boundary stabilizers as the bulk charge-conjugation membrane [Eq. (4)]. As we are acting on an open line, there will be nonsymmetric inputs to the gauging map at the end points; the resultant nonlocal stabilizers store the internal state of the non-Abelian defect pair (see Supplemental Material [57] for details).

$D(S_3)$ quantum double—The quantum double $D(S_3)$ is obtained by gauging the $\prod \mathcal{C}$ symmetry of the $\mathbb{Z}_3$ toric code [30]—upon gauging, the charge-conjugation defects become deconfined non-Abelian anyons. We immediately arrive at ribbon operators for these anyons by sandwiching the defect circuit with $\text{KW}_{\mathbb{Z}_2}^{\text{1D}}$. Other anyons in $D(S_3)$ originate from anyons of the $\mathbb{Z}_3$ toric code itself and so are created by gauging-map-decorated Abelian strings (see Table I).

We can implement the 1D gauging map via a sequential unitary circuit [53] or a finite-depth adaptive circuit (for solvable symmetry groups) [28,31,32]. The latter choice yields adaptive, efficient ribbons for non-Abelian excitations. Such circuits have been written down before for quantum doubles with solvable gauge group [31]—we emphasize, however, that our scheme applies more generally, as we will illustrate below. Moreover, it gives manifestly unitary ribbon operators for general non-Abelian anyons, which was left as an open question in Ref. [31].

General G gauging map—Our protocol to construct ribbons for the $\mathbb{Z}_3$ toric code and $D(S_3)$ can be generalized

TABLE I. Examples: We list exemplar topological orders whose ribbon operators are derived using the symmetry-membrane approach; further details can be found in Supplemental Material [57]. (a) All $D(S_3)$ anyons can be mapped back to anyons and defects of the $\mathbb{Z}_3$ toric code, as the two theories are related by gauging charge-conjugation symmetry. $D(S_3)$ ribbons are equivalent to $\mathbb{Z}_3$ ribbons decorated with $\text{KW}_{\mathbb{Z}_3}^{\text{ID}}$. (b),(c) Both $D(D_4)$ and $D(Q_8)$ are isomorphic to twisted quantum doubles [47–50]. We illustrate flux anyon ribbons corresponding to different $\mathbb{Z}_2$ symmetries: the legend indicates the constituent gates ($R = e^{i(\pi/4)Z}$).

Topological order	Duality map	Precursor phase	Ribbons
(a) $D(S_3)$	$\text{KW}_{\mathbb{Z}_2}$	$D(\mathbb{Z}_3)$	
(b) $D(D_4)$	$\text{KW}_{\mathbb{Z}_2^3}$	Type III $\mathbb{Z}_2^3$ SPT [58]	
(c) Quaternions [$D(Q_8)$]	$\text{KW}_{\mathbb{Z}_2^3}$	Type I + III $\mathbb{Z}_2^3$ SPT [59]	

to general quantum double $D(G)$ by using the finite-group G gauging map KW_G [22,32,60]. First, we define group-based Pauli operators for finite group G ,

$$L_g|h\rangle = |gh\rangle, \quad R_g|h\rangle = |h\bar{g}\rangle, \\ \mathcal{Z}_{\Gamma_{ij}}|h\rangle = \Gamma_{ij}(h)|h\rangle, \quad T_g|h\rangle = \delta_{g,h}|h\rangle, \quad (10)$$

where $g, h \in G$, $\Gamma \in \text{Rep}(G)$, and i, j are indices in the virtual representation space. KW_G gauges the $\prod R_g$ symmetry of the G paramagnet $\otimes_v (\sum_{g \in G} |g\rangle_v)$ to yield $D(G)$. It is defined by the following transformation rules:

$$L_g \xrightarrow{\text{KW}_G} A_g = R_g \begin{array}{c} L_g \\ R_g \end{array} L_g \quad \sum_k \mathcal{Z}_{\Gamma^{kj}} \xrightarrow{\text{KW}_G} \mathcal{Z}_{\Gamma^{ik}} \quad (11)$$

To create generic anyons and defects in $D(G)$, we first identify the corresponding precursor excitations in the G paramagnet. Charge anyons originate as charges of the G paramagnet, created by $\mathcal{Z}_{\Gamma^{(v)\dagger}} \cdot \mathcal{Z}_{\Gamma^{(v)'}}$ for some vertices v, v' —here, $(-\cdot-)$ indicates matrix multiplication in the virtual index space. After gauging, these charges live at the ends of open $\text{Rep}(G)$ 1-form symmetry operators [see Fig. 3(a)].

Flux anyons and symmetry defects are both related to automorphisms of the underlying group G [61]. When a flux g encircles some region, it transforms any flux h inside to $gh\bar{g}$. Conjugations by g generate $\text{Inn}(G)$, the “inner” automorphism group. Extrinsic symmetry defects correspond to “outer” automorphisms in $\text{Out}(G) = \text{Aut}(G)/\text{Inn}(G)$, which cannot be expressed as conjugation by any $g \in G$. For example, $\text{Out}(\mathbb{Z}_3) = \{1, \mathcal{C}\} \simeq \mathbb{Z}_2$.

Given an automorphism $\mathcal{O} \in \text{Aut}(G)$, KW_G maps $\prod_v \mathcal{O}^{(v)}$ into $\prod_e \mathcal{O}^{(e)}$ —therefore, an automorphism symmetry membrane is a sensible operator to push through KW_G . $\mathcal{O}$ can be decomposed in terms of Eq. (10),

$$\mathcal{O} = \sum_{g \in G} L_{\Omega(g)} T_g = \sum_{g \in G} R_{\Omega(\bar{g})} T_g, \quad (12)$$

where $\Omega(g) \equiv \mathcal{O}(g)\bar{g}$. Combining Eqs. (11) and (12) yields

$$\text{KW}_G \cdot \mathcal{O} = \sum_g A_{\Omega(g)} \cdot \text{KW}_G \cdot T_g \\ \mathcal{O} \cdot \text{KW}_G = \sum_{g,h} R_{\Omega(\bar{g})} L_{\Omega(h)} \cdot \text{KW}_G \cdot \frac{T_g}{T_h} \quad (13)$$

Thus, a finite membrane of $\mathcal{O}^{(e)}$ is decorated by a boundary circuit of control gates when pushed through KW_G

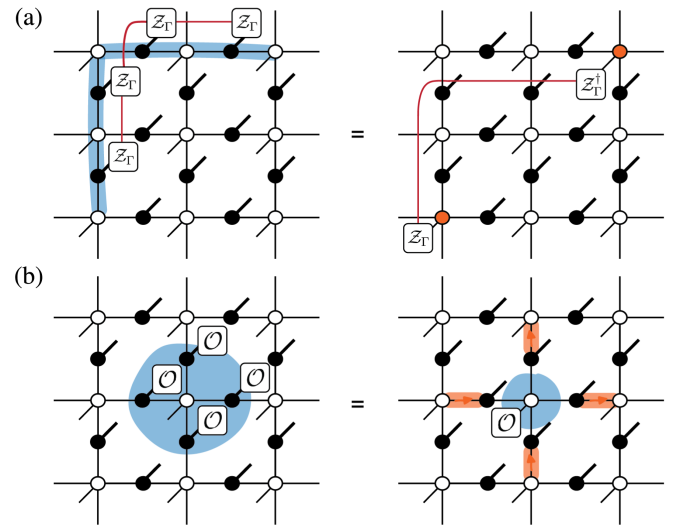


FIG. 3. Anyons and defects from the G gauging map: (a) A charge anyon ribbon maps to a pair of charges on top of the paramagnet. (b) The automorphism symmetry membrane on edges pushes through the gauging map the vertices, now decorated with a boundary circuit (orange lines). If the initial state is a paramagnet, the right-hand membrane disappears, leaving only the boundary circuit.

[see Fig. 3(b)]. An open defect or flux ribbon is constructed by truncating this circuit to an open line. The pregauged degrees of freedom acting as controls can be accessed by applying KW_G^{ID} along path γ , realized unitarily by $\prod_{i \in \gamma} CL^{(i \rightarrow i+1)}$, where $CL|h, g\rangle = |h, hg\rangle$.

Twisted quantum doubles—So far, we have leveraged the duality between the trivial G paramagnet and $D(G)$. Nontrivial symmetry-protected topological (SPT) phases are dual to twisted quantum doubles [47–50]—using this duality, we can construct ribbon operators for a broader set of topological phases. In particular, we obtain simple circuits for doubled semion order [62–64], $D(D_4)$, and $D(Q_8)$ ribbons [65]—see Table I. We provide detailed derivations for all three examples in Supplemental Material [57].

Here, we briefly discuss $D(D_4)$, which has recently been experimentally realized [36]. $D(D_4)$ is isomorphic to a $\mathbb{Z}_2^3$ twisted quantum double, obtained from gauging the type-III $\mathbb{Z}_2^3$ SPT [49,58]. This SPT can be written as $\prod_{\Delta} \text{CCZ } |+\rangle$ on the triangular lattice. The non-Abelian fluxes of $D(D_4)$ correspond to finite membranes of the $\prod X$ sublattice symmetries of the SPT—such membranes fractionalize on the SPT degrees of freedom to a product of CZ gates spanning different sublattices. When we push these gates through the gauging map, they will become nonlocal operators, which can be written as linear-depth circuits (see Table I), making the corresponding flux anyons non-Abelian.

Discussion—We have presented a method for deriving anyon and (outer automorphism) defect ribbons for $D(G)$. We obtain ribbon circuits spanning both vertices and edges by pushing precursor operators through the gauging map; applying a further 1D gauging map ensures the circuits are supported only on gauged degrees of freedom. Our protocol yields unitary expressions for anyons of generic $D(G)$ —additionally, we have introduced completely new ribbons for symmetry defects. In the case where only Abelian or solvable symmetry groups are considered, our method gives a simple way to construct finite-depth ribbons for non-Abelian excitations, by replacing all gauging steps with their efficient, measurement-based realizations. This is a powerful practical tool for constructing ribbon circuits for near-term quantum devices.

Non-Abelian excitations provide a possible route to universal, topologically protected quantum computing [21,66]—as non-Abelian phases and excitations become more accessible to quantum platforms [36,51,67–70], it is necessary to find concrete circuit realizations for the creation of any desired topological excitation. Scalability is also a key consideration given finite coherence times on near-term devices; while unitary non-Abelian ribbon operators cannot be finite depth [31,71], we provide a framework for constructing efficient nonunitary ribbons, thereby extending the work of Ref. [31]. Recently, finite-depth unitary circuits forgoing strict spatial locality

have been proposed to prepare topologically ordered states [72]—such methods may be useful in our framework as well to develop efficient unitary ribbons. An interesting direction would be to evaluate and improve the practicality of our circuits for fault-tolerant computing.

We emphasize that our ideas are not restricted to conventional quantum doubles. Twisted quantum double ribbons can be derived by using a nontrivial SPT as input to the gauging map: symmetry fractionalization on the input SPT must also be considered. Using the higher-dimensional gauging map, we can generalize to topological orders in $d \geq 2$. More exotic symmetry defects like the electromagnetic duality can also be accommodated in our framework, given a lattice realization of the symmetry.

Recent progress has been made on understanding string-net states [63] as gauge theories [73,74]; constructing a string-net gauging map could allow us to generalize our results to generic string-net states. We note that unitary, linear-depth, non-Abelian string operators for string-net states have been introduced in Refs. [75] and [70]—it would be interesting to compare with the ribbons obtained in our framework.

Acknowledgments—We thank the authors of Ref. [76] for informing us of their concurrent work, which discusses efficiently pairing up anyons using adaptive circuits. A. L. and C. F. B. L. acknowledge support from the National Science Foundation Graduate Research Fellowship Program (NSF GRFP). This work is in part supported by the DARPA MeasQuIT program. N. T. is supported by the Walter Burke Institute for Theoretical Physics at Caltech. A. V. and R. V. are supported by the Simons Collaboration on Ultra-Quantum Matter, which is a grant from the Simons Foundation (No. 618615, A. V.).

Data availability—No data were created or analyzed in this Letter.

-
- [1] H. A. Kramers and G. H. Wannier, *Phys. Rev.* **60**, 252 (1941).
 - [2] F. J. Wegner, *J. Math. Phys. (N.Y.)* **12**, 2259 (1971).
 - [3] J. B. Kogut, *Rev. Mod. Phys.* **51**, 659 (1979).
 - [4] S. Trebst, P. Werner, M. Troyer, K. Shtengel, and C. Nayak, *Phys. Rev. Lett.* **98**, 070602 (2007).
 - [5] I. S. Tupitsyn, A. Kitaev, N. V. Prokof'ev, and P. C. E. Stamp, *Phys. Rev. B* **82**, 085114 (2010).
 - [6] E. Cobanera, G. Ortiz, and Z. Nussinov, *Adv. Phys.* **60**, 679 (2011).
 - [7] J. Haegeman, K. Van Acoleyen, N. Schuch, J. I. Cirac, and F. Verstraete, *Phys. Rev. X* **5**, 011024 (2015).
 - [8] S. Vijay, J. Haah, and L. Fu, *Phys. Rev. B* **94**, 235157 (2016).
 - [9] B. Yoshida, *Ann. Phys. (Amsterdam)* **377**, 387 (2017).
 - [10] D. J. Williamson, N. Bultinck, and F. Verstraete, *arXiv:1711.07982*.
 - [11] A. Kubica and B. Yoshida, *arXiv:1805.01836*.

- [12] W. Shirley, K. Slagle, and X. Chen, *SciPost Phys.* **6**, 041 (2019).
- [13] N. Tantivasadakarn, *Phys. Rev. Res.* **2**, 023353 (2020).
- [14] N. Tantivasadakarn and S. Vijay, *Phys. Rev. B* **101**, 165143 (2020).
- [15] D. Radicevic, *arXiv:1910.06336*.
- [16] K. Dolev, V. Calvera, S. S. Cree, and D. J. Williamson, *J. High Energy Phys.* **05** (2022) 158.
- [17] B. Rayhaun and D. Williamson, *SciPost Phys.* **15**, 017 (2023).
- [18] T. Okuda, A. P. Mana, and H. Sukeno, *SciPost Phys.* **17**, 113 (2024).
- [19] N. Seiberg, S. Seifnashri, and S.-H. Shao, *SciPost Phys.* **16**, 154 (2024).
- [20] N. Seiberg and S.-H. Shao, *SciPost Phys.* **16**, 064 (2024).
- [21] A. Yu. Kitaev, *Ann. Phys. (Amsterdam)* **303**, 2 (2003).
- [22] C. G. Brell, *New J. Phys.* **17**, 023029 (2015).
- [23] J. M. Leinaas and J. Myrheim, *Nuovo Cimento Soc. Ital. Fis.* **37B**, 1 (1977).
- [24] F. Wilczek, *Phys. Rev. Lett.* **49**, 957 (1982).
- [25] G. A. Goldin, R. Menikoff, and D. H. Sharp, *Phys. Rev. Lett.* **54**, 603 (1985).
- [26] J. Fröhlich, F. Gabbiani, and P.-A. Marchetti, Braid statistics in three-dimensional local quantum theory, in *Physics, Geometry and Topology*, edited by H. C. Lee (Springer, Boston, MA, 1990), pp. 15–79.
- [27] X. G. Wen, *Phys. Rev. Lett.* **66**, 802 (1991).
- [28] N. Tantivasadakarn, R. Thorngren, A. Vishwanath, and R. Verresen, *Phys. Rev. X* **14**, 021040 (2024).
- [29] N. Tantivasadakarn, R. Verresen, and A. Vishwanath, *Phys. Rev. Lett.* **131**, 060405 (2023).
- [30] R. Verresen, N. Tantivasadakarn, and A. Vishwanath, *arxiv:2112.03061*.
- [31] S. Bravyi, I. Kim, A. Kliesch, and R. Koenig, *arxiv:2205.01933*.
- [32] N. Tantivasadakarn, A. Vishwanath, and R. Verresen, *PRX Quantum* **4**, 020339 (2023).
- [33] Y. Li, H. Sukeno, A. P. Mana, H. P. Nautrup, and T.-C. Wei, *Phys. Rev. B* **108**, 115144 (2023).
- [34] H. Sukeno and T.-C. Wei, *Phys. Rev. A* **109**, 042611 (2024).
- [35] Y. Li, M. Litvinov, and T.-C. Wei, *arXiv:2310.17740*.
- [36] M. Iqbal, N. Tantivasadakarn, R. Verresen, S. L. Campbell, J. M. Dreiling, C. Figgatt, J. P. Gaebler, J. Johansen, M. Mills, S. A. Moses, J. M. Pino, A. Ransford, M. Rowe, P. Siegfried, R. P. Stutz, M. Foss-Feig, A. Vishwanath, and H. Dreyer, *Nature (London)* **626**, 505 (2024).
- [37] H. Bombin and M. A. Martin-Delgado, *Phys. Rev. B* **78**, 115421 (2008).
- [38] B. Yan, P. Chen, and S. X. Cui, *J. Phys. A* **55**, 185201 (2022).
- [39] M. Barkeshli and X.-G. Wen, *Phys. Rev. B* **81**, 045323 (2010).
- [40] H. Bombin, *Phys. Rev. Lett.* **105**, 030403 (2010).
- [41] Y.-Z. You and X.-G. Wen, *Phys. Rev. B* **86**, 161107(R) (2012).
- [42] M. Barkeshli, C.-M. Jian, and X.-L. Qi, *Phys. Rev. B* **87**, 045130 (2013).
- [43] Y.-Z. You, C.-M. Jian, and X.-G. Wen, *Phys. Rev. B* **87**, 045106 (2013).
- [44] M. Barkeshli, C.-M. Jian, and X.-L. Qi, *Phys. Rev. B* **88**, 235103 (2013).
- [45] M. Barkeshli, C.-M. Jian, and X.-L. Qi, *Phys. Rev. B* **88**, 241103(R) (2013).
- [46] M. Barkeshli, P. Bonderson, M. Cheng, and Z. Wang, *Phys. Rev. B* **100**, 115147 (2019).
- [47] R. Dijkgraaf and E. Witten, *Commun. Math. Phys.* **129**, 393 (1990).
- [48] P. Roche, V. Pasquier, and R. Dijkgraaf, *Nucl. Phys. B, Proc. Suppl.* **18**, 60 (1990).
- [49] M. d. W. Propitius, Topological interactions in broken gauge theories, Ph.D. thesis, Institute for Theoretical Physics Amsterdam, 1995.
- [50] Y. Hu, Y. Wan, and Y.-S. Wu, *Phys. Rev. B* **87**, 125114 (2013).
- [51] M. Iqbal, A. Lyons, C. F. B. Lo, N. Tantivasadakarn, J. Dreiling, C. Foltz, T. M. Gatterman, D. Gresh, N. Hewitt, C. A. Holliman, J. Johansen, B. Neyenhuis, Y. Matsuoka, M. Mills, S. A. Moses, P. Siegfried, A. Vishwanath, R. Verresen, and H. Dreyer, *Nat. Commun.* **16**, 6301 (2025).
- [52] We are making a choice of lattice orientation where horizontal edges point right and vertical edges point upward. This convention is constant throughout.
- [53] X. Chen, A. Dua, M. Hermele, D. T. Stephen, N. Tantivasadakarn, R. Vanhove, and J.-Y. Zhao, *Phys. Rev. B* **109**, 075116 (2024).
- [54] The measurement-based circuit involves explicitly introducing new degrees of freedom on the vertices to map from the ferromagnetic phase to the paramagnetic, while the unitary circuit maps between wave functions on the same degrees of freedom.
- [55] K. Roumpedakis, S. Seifnashri, and S.-H. Shao, *Commun. Math. Phys.* **401**, 3043 (2023).
- [56] For a 2D model, a 1-form symmetry lives on deformable 1D submanifolds; the logical operators of the toric code are an example.
- [57] See Supplemental Material at <http://link.aps.org/supplemental/10.1103/q2c6-nd6r> for more detailed derivations of results stated in the main text, as well as additional examples.
- [58] B. Yoshida, *Phys. Rev. B* **93**, 155131 (2016).
- [59] M. de Wild Propitius and F. A. Bais, in *CRM-CAP Summer School on Particles and Fields '94* (Springer, New York, 1995), pp. 353–439.
- [60] C. Feichsin, N. Tantivasadakarn, and V. V. Albert, *Phys. Rev. X* **15**, 011058 (2025).
- [61] Alternatively, fluxes can be traced back to charges on top of a $\text{Rep}(G)$ paramagnet. See Ref. [57] for more information, including a derivation of the usual flux ribbon operators using the properties of the $\text{Rep}(G)$ gauging map.
- [62] M. Freedman, C. Nayak, K. Shtengel, K. Walker, and Z. Wang, *Ann. Phys. (Amsterdam)* **310**, 428 (2004).
- [63] M. A. Levin and X.-G. Wen, *Phys. Rev. B* **71**, 045110 (2005).
- [64] M. Levin and Z.-C. Gu, *Phys. Rev. B* **86**, 115109 (2012).
- [65] While the latter two examples are conventional quantum doubles for the non-Abelian groups D_4 and Q_8 , they are each isomorphic to a twisted quantum double of $\mathbb{Z}_2^3$.
- [66] C. Nayak, S. H. Simon, A. Stern, M. Freedman, and S. Das Sarma, *Rev. Mod. Phys.* **80**, 1083 (2008).

- [67] M. Iqbal, N. Tantivasadakarn, T. M. Gatterman, J. A. Gerber, K. Gilmore, D. Gresh, A. Hankin, N. Hewitt, C. V. Horst, M. Matheny *et al.*, [Commun. Phys. **7**, 205 \(2024\)](#).
- [68] T. I. Andersen, Y. D. Lensky, K. Kechedzhi, I. Drozdov, A. Bengtsson, S. Hong, A. Morvan, X. Mi, A. Opremcak, R. Acharya *et al.*, [Nature \(London\) **618**, 264 \(2023\)](#).
- [69] S. Xu, Z.-Z. Sun, K. Wang, H. Li, Z. Zhu, H. Dong, J. Deng, X. Zhang, J. Chen, Y. Wu *et al.*, [Nat. Phys. **20**, 1469 \(2024\)](#).
- [70] Z. K. Mineev, K. Najafi, S. Majumder, J. Wang, A. Stern, E.-A. Kim, C.-M. Jian, and G. Zhu, [Nat. Commun. **16**, 6225 \(2025\)](#).
- [71] B. Shi, [Phys. Rev. Res. **1**, 033048 \(2019\)](#).
- [72] Y. Sheffer, E. Berg, and A. Stern, [Phys. Rev. B **111**, 115160 \(2025\)](#).
- [73] K. Kawagoe, C. Jones, S. Sanford, D. Green, and D. Penneys, [Commun. Math. Phys. **405**, 266 \(2024\)](#).
- [74] Y. Zhao and Y. Wan, [arXiv:2408.02664](#).
- [75] Y.-J. Liu, K. Shtengel, A. Smith, and F. Pollmann, [PRX Quantum **3**, 040315 \(2022\)](#).
- [76] Y. Ren, N. Tantivasadakarn, and D. J. Williamson, [Phys. Rev. X **15**, 031060 \(2025\)](#).